

Primary 1-3

INGREDIENT LAB ACTIVITY SYSTEM

IL-101 to IL-200

OBSERVE -> CLASSIFY -> MEASURE -> COMPARE -> RECORD

Purpose	Build food and ingredient literacy by connecting plant structure, observation, measurement, post-harvest change, quality and product-development thinking.
Ingredient set	Hibiscus, ginger, turmeric, pineapple, citrus, mint, moringa, watermelon, tomato and maize.
Core rule	The Ingredient Lab is an educational investigation environment. It does not require tasting, medical claims or uncontrolled food processing.
Connections	Link each activity to GrowMeal vocabulary, journals, practical cards, Farm-to-Food activities, missions and assessment.

INGREDIENT LAB - HIBISCUS | PLANT PART: CALYX

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-101 - HIBISCUS - IDENTITY & PLANT PART

Plant part	Calyx
Learning objective	Observe and classify hibiscus as a calyx, then use simple measurement or comparison to investigate identity & plant part.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-102 - HIBISCUS - EXTERNAL STRUCTURE

Plant part	Calyx
Learning objective	Observe and classify hibiscus as a calyx, then use simple measurement or comparison to investigate external structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-103 - HIBISCUS - INTERNAL STRUCTURE

Plant part	Calyx
Learning objective	Observe and classify hibiscus as a calyx, then use simple measurement or comparison to investigate internal structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-104 - HIBISCUS - COLOUR & TEXTURE

Plant part	Calyx
Learning objective	Observe and classify hibiscus as a calyx, then use simple measurement or comparison to investigate colour & texture.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-105 - HIBISCUS - MASS & SIZE

Plant part	Calyx
Learning objective	Observe and classify hibiscus as a calyx, then use simple measurement or comparison to investigate mass & size.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-106 - HIBISCUS - WATER/CHANGE OBSERVATION

Plant part	Calyx
Learning objective	Observe and classify hibiscus as a calyx, then use simple measurement or comparison to investigate water/change observation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-107 - HIBISCUS - SORTING & QUALITY

Plant part	Calyx
Learning objective	Observe and classify hibiscus as a calyx, then use simple measurement or comparison to investigate sorting & quality.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-108 - HIBISCUS - POST-HARVEST CHANGE

Plant part	Calyx
Learning objective	Observe and classify hibiscus as a calyx, then use simple measurement or comparison to investigate post-harvest change.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-109 - HIBISCUS - INGREDIENT-TO-FOOD PATHWAY

Plant part	Calyx
Learning objective	Observe and classify hibiscus as a calyx, then use simple measurement or comparison to investigate ingredient-to-food pathway.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-110 - HIBISCUS - INVESTIGATION CHALLENGE

Plant part	Calyx
Learning objective	Observe and classify hibiscus as a calyx, then use simple measurement or comparison to investigate investigation challenge.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - GINGER | PLANT PART: RHIZOME

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-111 - GINGER - IDENTITY & PLANT PART

Plant part	Rhizome
Learning objective	Observe and classify ginger as a rhizome, then use simple measurement or comparison to investigate identity & plant part.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-112 - GINGER - EXTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Observe and classify ginger as a rhizome, then use simple measurement or comparison to investigate external structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-113 - GINGER - INTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Observe and classify ginger as a rhizome, then use simple measurement or comparison to investigate internal structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-114 - GINGER - COLOUR & TEXTURE

Plant part	Rhizome
Learning objective	Observe and classify ginger as a rhizome, then use simple measurement or comparison to investigate colour & texture.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-115 - GINGER - MASS & SIZE

Plant part	Rhizome
Learning objective	Observe and classify ginger as a rhizome, then use simple measurement or comparison to investigate mass & size.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-116 - GINGER - WATER/CHANGE OBSERVATION

Plant part	Rhizome
Learning objective	Observe and classify ginger as a rhizome, then use simple measurement or comparison to investigate water/change observation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-117 - GINGER - SORTING & QUALITY

Plant part	Rhizome
Learning objective	Observe and classify ginger as a rhizome, then use simple measurement or comparison to investigate sorting & quality.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-118 - GINGER - POST-HARVEST CHANGE

Plant part	Rhizome
Learning objective	Observe and classify ginger as a rhizome, then use simple measurement or comparison to investigate post-harvest change.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-119 - GINGER - INGREDIENT-TO-FOOD PATHWAY

Plant part	Rhizome
Learning objective	Observe and classify ginger as a rhizome, then use simple measurement or comparison to investigate ingredient-to-food pathway.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-120 - GINGER - INVESTIGATION CHALLENGE

Plant part	Rhizome
Learning objective	Observe and classify ginger as a rhizome, then use simple measurement or comparison to investigate investigation challenge.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
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AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - TURMERIC | PLANT PART: RHIZOME

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-121 - TURMERIC - IDENTITY & PLANT PART

Plant part	Rhizome
Learning objective	Observe and classify turmeric as a rhizome, then use simple measurement or comparison to investigate identity & plant part.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-122 - TURMERIC - EXTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Observe and classify turmeric as a rhizome, then use simple measurement or comparison to investigate external structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-123 - TURMERIC - INTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Observe and classify turmeric as a rhizome, then use simple measurement or comparison to investigate internal structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
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AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-124 - TURMERIC - COLOUR & TEXTURE

Plant part	Rhizome
Learning objective	Observe and classify turmeric as a rhizome, then use simple measurement or comparison to investigate colour & texture.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-125 - TURMERIC - MASS & SIZE

Plant part	Rhizome
Learning objective	Observe and classify turmeric as a rhizome, then use simple measurement or comparison to investigate mass & size.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-126 - TURMERIC - WATER/CHANGE OBSERVATION

Plant part	Rhizome
Learning objective	Observe and classify turmeric as a rhizome, then use simple measurement or comparison to investigate water/change observation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-127 - TURMERIC - SORTING & QUALITY

Plant part	Rhizome
Learning objective	Observe and classify turmeric as a rhizome, then use simple measurement or comparison to investigate sorting & quality.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-128 - TURMERIC - POST-HARVEST CHANGE

Plant part	Rhizome
Learning objective	Observe and classify turmeric as a rhizome, then use simple measurement or comparison to investigate post-harvest change.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-129 - TURMERIC - INGREDIENT-TO-FOOD PATHWAY

Plant part	Rhizome
Learning objective	Observe and classify turmeric as a rhizome, then use simple measurement or comparison to investigate ingredient-to-food pathway.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-130 - TURMERIC - INVESTIGATION CHALLENGE

Plant part	Rhizome
Learning objective	Observe and classify turmeric as a rhizome, then use simple measurement or comparison to investigate investigation challenge.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - PINEAPPLE | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-131 - PINEAPPLE - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Observe and classify pineapple as a fruit, then use simple measurement or comparison to investigate identity & plant part.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-132 - PINEAPPLE - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Observe and classify pineapple as a fruit, then use simple measurement or comparison to investigate external structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-133 - PINEAPPLE - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Observe and classify pineapple as a fruit, then use simple measurement or comparison to investigate internal structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-134 - PINEAPPLE - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Observe and classify pineapple as a fruit, then use simple measurement or comparison to investigate colour & texture.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-135 - PINEAPPLE - MASS & SIZE

Plant part	Fruit
Learning objective	Observe and classify pineapple as a fruit, then use simple measurement or comparison to investigate mass & size.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-136 - PINEAPPLE - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Observe and classify pineapple as a fruit, then use simple measurement or comparison to investigate water/change observation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-137 - PINEAPPLE - SORTING & QUALITY

Plant part	Fruit
Learning objective	Observe and classify pineapple as a fruit, then use simple measurement or comparison to investigate sorting & quality.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-138 - PINEAPPLE - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Observe and classify pineapple as a fruit, then use simple measurement or comparison to investigate post-harvest change.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-139 - PINEAPPLE - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Observe and classify pineapple as a fruit, then use simple measurement or comparison to investigate ingredient-to-food pathway.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-140 - PINEAPPLE - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Observe and classify pineapple as a fruit, then use simple measurement or comparison to investigate investigation challenge.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - CITRUS | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-141 - CITRUS - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Observe and classify citrus as a fruit, then use simple measurement or comparison to investigate identity & plant part.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-142 - CITRUS - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Observe and classify citrus as a fruit, then use simple measurement or comparison to investigate external structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-143 - CITRUS - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Observe and classify citrus as a fruit, then use simple measurement or comparison to investigate internal structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-144 - CITRUS - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Observe and classify citrus as a fruit, then use simple measurement or comparison to investigate colour & texture.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-145 - CITRUS - MASS & SIZE

Plant part	Fruit
Learning objective	Observe and classify citrus as a fruit, then use simple measurement or comparison to investigate mass & size.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-146 - CITRUS - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Observe and classify citrus as a fruit, then use simple measurement or comparison to investigate water/change observation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-147 - CITRUS - SORTING & QUALITY

Plant part	Fruit
Learning objective	Observe and classify citrus as a fruit, then use simple measurement or comparison to investigate sorting & quality.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-148 - CITRUS - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Observe and classify citrus as a fruit, then use simple measurement or comparison to investigate post-harvest change.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-149 - CITRUS - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Observe and classify citrus as a fruit, then use simple measurement or comparison to investigate ingredient-to-food pathway.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-150 - CITRUS - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Observe and classify citrus as a fruit, then use simple measurement or comparison to investigate investigation challenge.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MINT | PLANT PART: LEAF

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-151 - MINT - IDENTITY & PLANT PART

Plant part	Leaf
Learning objective	Observe and classify mint as a leaf, then use simple measurement or comparison to investigate identity & plant part.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-152 - MINT - EXTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Observe and classify mint as a leaf, then use simple measurement or comparison to investigate external structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-153 - MINT - INTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Observe and classify mint as a leaf, then use simple measurement or comparison to investigate internal structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-154 - MINT - COLOUR & TEXTURE

Plant part	Leaf
Learning objective	Observe and classify mint as a leaf, then use simple measurement or comparison to investigate colour & texture.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-155 - MINT - MASS & SIZE

Plant part	Leaf
Learning objective	Observe and classify mint as a leaf, then use simple measurement or comparison to investigate mass & size.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-156 - MINT - WATER/CHANGE OBSERVATION

Plant part	Leaf
Learning objective	Observe and classify mint as a leaf, then use simple measurement or comparison to investigate water/change observation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-157 - MINT - SORTING & QUALITY

Plant part	Leaf
Learning objective	Observe and classify mint as a leaf, then use simple measurement or comparison to investigate sorting & quality.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-158 - MINT - POST-HARVEST CHANGE

Plant part	Leaf
Learning objective	Observe and classify mint as a leaf, then use simple measurement or comparison to investigate post-harvest change.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-159 - MINT - INGREDIENT-TO-FOOD PATHWAY

Plant part	Leaf
Learning objective	Observe and classify mint as a leaf, then use simple measurement or comparison to investigate ingredient-to-food pathway.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-160 - MINT - INVESTIGATION CHALLENGE

Plant part	Leaf
Learning objective	Observe and classify mint as a leaf, then use simple measurement or comparison to investigate investigation challenge.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MORINGA | PLANT PART: LEAF

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-161 - MORINGA - IDENTITY & PLANT PART

Plant part	Leaf
Learning objective	Observe and classify moringa as a leaf, then use simple measurement or comparison to investigate identity & plant part.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-162 - MORINGA - EXTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Observe and classify moringa as a leaf, then use simple measurement or comparison to investigate external structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-163 - MORINGA - INTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Observe and classify moringa as a leaf, then use simple measurement or comparison to investigate internal structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-164 - MORINGA - COLOUR & TEXTURE

Plant part	Leaf
Learning objective	Observe and classify moringa as a leaf, then use simple measurement or comparison to investigate colour & texture.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-165 - MORINGA - MASS & SIZE

Plant part	Leaf
Learning objective	Observe and classify moringa as a leaf, then use simple measurement or comparison to investigate mass & size.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-166 - MORINGA - WATER/CHANGE OBSERVATION

Plant part	Leaf
Learning objective	Observe and classify moringa as a leaf, then use simple measurement or comparison to investigate water/change observation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-167 - MORINGA - SORTING & QUALITY

Plant part	Leaf
Learning objective	Observe and classify moringa as a leaf, then use simple measurement or comparison to investigate sorting & quality.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-168 - MORINGA - POST-HARVEST CHANGE

Plant part	Leaf
Learning objective	Observe and classify moringa as a leaf, then use simple measurement or comparison to investigate post-harvest change.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-169 - MORINGA - INGREDIENT-TO-FOOD PATHWAY

Plant part	Leaf
Learning objective	Observe and classify moringa as a leaf, then use simple measurement or comparison to investigate ingredient-to-food pathway.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-170 - MORINGA - INVESTIGATION CHALLENGE

Plant part	Leaf
Learning objective	Observe and classify moringa as a leaf, then use simple measurement or comparison to investigate investigation challenge.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - WATERMELON | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-171 - WATERMELON - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Observe and classify watermelon as a fruit, then use simple measurement or comparison to investigate identity & plant part.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-172 - WATERMELON - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Observe and classify watermelon as a fruit, then use simple measurement or comparison to investigate external structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-173 - WATERMELON - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Observe and classify watermelon as a fruit, then use simple measurement or comparison to investigate internal structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-174 - WATERMELON - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Observe and classify watermelon as a fruit, then use simple measurement or comparison to investigate colour & texture.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-175 - WATERMELON - MASS & SIZE

Plant part	Fruit
Learning objective	Observe and classify watermelon as a fruit, then use simple measurement or comparison to investigate mass & size.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-176 - WATERMELON - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Observe and classify watermelon as a fruit, then use simple measurement or comparison to investigate water/change observation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-177 - WATERMELON - SORTING & QUALITY

Plant part	Fruit
Learning objective	Observe and classify watermelon as a fruit, then use simple measurement or comparison to investigate sorting & quality.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-178 - WATERMELON - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Observe and classify watermelon as a fruit, then use simple measurement or comparison to investigate post-harvest change.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-179 - WATERMELON - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Observe and classify watermelon as a fruit, then use simple measurement or comparison to investigate ingredient-to-food pathway.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-180 - WATERMELON - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Observe and classify watermelon as a fruit, then use simple measurement or comparison to investigate investigation challenge.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - TOMATO | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-181 - TOMATO - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Observe and classify tomato as a fruit, then use simple measurement or comparison to investigate identity & plant part.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-182 - TOMATO - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Observe and classify tomato as a fruit, then use simple measurement or comparison to investigate external structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-183 - TOMATO - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Observe and classify tomato as a fruit, then use simple measurement or comparison to investigate internal structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-184 - TOMATO - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Observe and classify tomato as a fruit, then use simple measurement or comparison to investigate colour & texture.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-185 - TOMATO - MASS & SIZE

Plant part	Fruit
Learning objective	Observe and classify tomato as a fruit, then use simple measurement or comparison to investigate mass & size.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-186 - TOMATO - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Observe and classify tomato as a fruit, then use simple measurement or comparison to investigate water/change observation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-187 - TOMATO - SORTING & QUALITY

Plant part	Fruit
Learning objective	Observe and classify tomato as a fruit, then use simple measurement or comparison to investigate sorting & quality.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-188 - TOMATO - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Observe and classify tomato as a fruit, then use simple measurement or comparison to investigate post-harvest change.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-189 - TOMATO - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Observe and classify tomato as a fruit, then use simple measurement or comparison to investigate ingredient-to-food pathway.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-190 - TOMATO - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Observe and classify tomato as a fruit, then use simple measurement or comparison to investigate investigation challenge.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MAIZE | PLANT PART: SEED/GRAIN

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-191 - MAIZE - IDENTITY & PLANT PART

Plant part	Seed/Grain
Learning objective	Observe and classify maize as a seed/grain, then use simple measurement or comparison to investigate identity & plant part.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-192 - MAIZE - EXTERNAL STRUCTURE

Plant part	Seed/Grain
Learning objective	Observe and classify maize as a seed/grain, then use simple measurement or comparison to investigate external structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-193 - MAIZE - INTERNAL STRUCTURE

Plant part	Seed/Grain
Learning objective	Observe and classify maize as a seed/grain, then use simple measurement or comparison to investigate internal structure.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-194 - MAIZE - COLOUR & TEXTURE

Plant part	Seed/Grain
Learning objective	Observe and classify maize as a seed/grain, then use simple measurement or comparison to investigate colour & texture.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-195 - MAIZE - MASS & SIZE

Plant part	Seed/Grain
Learning objective	Observe and classify maize as a seed/grain, then use simple measurement or comparison to investigate mass & size.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-196 - MAIZE - WATER/CHANGE OBSERVATION

Plant part	Seed/Grain
Learning objective	Observe and classify maize as a seed/grain, then use simple measurement or comparison to investigate water/change observation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-197 - MAIZE - SORTING & QUALITY

Plant part	Seed/Grain
Learning objective	Observe and classify maize as a seed/grain, then use simple measurement or comparison to investigate sorting & quality.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-198 - MAIZE - POST-HARVEST CHANGE

Plant part	Seed/Grain
Learning objective	Observe and classify maize as a seed/grain, then use simple measurement or comparison to investigate post-harvest change.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-199 - MAIZE - INGREDIENT-TO-FOOD PATHWAY

Plant part	Seed/Grain
Learning objective	Observe and classify maize as a seed/grain, then use simple measurement or comparison to investigate ingredient-to-food pathway.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-200 - MAIZE - INVESTIGATION CHALLENGE

Plant part	Seed/Grain
Learning objective	Observe and classify maize as a seed/grain, then use simple measurement or comparison to investigate investigation challenge.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Observe teacher-approved samples, record visible properties, make a count or simple measurement, compare two examples, and write/draw the result.
Evidence to record	Labeled observation, simple table, measurement, comparison sentence or diagram.
Questions	What did you observe? Which property can we measure? How are the samples similar or different?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?